

Debating the Trolley Problem: A Multi-LLM Framework for Assessing Moral Reasoning and Persona-Driven Ethical Judgment

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Abstract

We investigate whether large language models conditioned on real conversation transcripts of specific individuals genuinely reason like those people, or merely sound like them. Using the trolley problem as a controlled probe of moral reasoning, we recruit three participants screened with the Oxford Utilitarianism Scale to span utilitarian, deontological, and intermediate moral leanings. Each participant sits for an interview discussing their values and reasoning on unrelated moral scenarios; these transcripts become persona-conditioning material for three frontier models from three labs, GPT-5.5 Pro, Claude Opus 4.7, and Gemini 3.1 Pro. Each model produces solo verdicts on three locked trolley variants (Standard Switch, Footbridge, Loved One) for each persona, yielding 27 generations, followed by nine within-model debates among persona-conditioned instances. Participants then rate their agreement with the conditioned-model arguments on a 1–5 scale, and the participant self-report serves as the verdict-match ground truth. By comparing persona-conditioned models across labs and combining verdict-match with participant agreement ratings, we assess whether persona conditioning shifts how a model reasons or only how it sounds. Our contribution combines cross-model debate, organic persona grounding, and human-in-the-loop fidelity evaluation in a single framework.

1. Introduction

As large language models are increasingly deployed as conversational agents and decision-support tools, a pressing question for artificial social intelligence is whether they can authentically represent the moral perspectives of specific individuals, not merely produce socially plausible text. Persona-conditioned LLMs are already used to simulate users in survey research, generate synthetic focus groups, and roleplay specific personalities, yet it remains unclear whether such conditioning shifts the underlying structure of a model’s reasoning or only its surface style. The trolley problem offers a uniquely controlled probe: by varying one dimension at a time (personal force, partiality, act-versus-omission) while holding the utilitarian calculus constant, its variants isolate the specific moral commitments driving a verdict, so a genuinely persona-conditioned model should track an individual’s idiosyncratic pattern *across* variants rather than echo the model’s defaults.

This is hard for three reasons. Frontier LLMs exhibit strong default moral tendencies that vary systematically by lab [3] and amplify cognitive biases rela-

tive to human subjects [2], priors that may dominate weak persona signals. Existing persona-assignment studies rely on synthetic backstories, demographic labels, or political identities [4, 6] rather than organic conversation data from specific people. And no prior work distinguishes whether a model captures a person’s reasoning or simply happens to agree with them, conflating fidelity with verdict alignment.

We address these gaps with a framework combining three locked trolley variants, three frontier models from three labs, three real participants screened to span moral leanings, and a participant-in-the-loop evaluation that separates agreement from fidelity. Each persona-conditioned model produces solo verdicts and then debates the other personas within its own model family, letting us track stance shifts and reasoning style across rounds. The rest of this paper is organized as follows: Section 2 reviews related work; Section 3 describes participants, interviews, and trolley variants; Section 4 details the generation and debate pipeline; Section 5 presents results; Section 6 discusses implications; Section 7 covers ethics and the model card; Section 8 concludes.

2. Related Work

The trolley problem has long served as a canonical probe of the tension between utilitarian and deontological reasoning, with the *Moral Machine* experiment of Awad et al. [1] establishing the empirical baseline of human moral judgments (over 40 million decisions across cultures) against which contemporary LLMs are now routinely compared.

2.1. LLM Moral Bias and Multi-Agent Behaviour

A growing body of work uses moral dilemmas to audit the inherent biases of LLMs. Xu et al. [8] evaluated nine LLMs across 50,400 dilemma trials involving protected attributes, finding that open-source models favour marginalised groups while proprietary closed-source models favour mainstream groups. Hatemo et al. [4] introduced demographic prompts (age, gender, nationality) into trolley variants across LLaMA, Mistral, and Qwen, identifying nationality as the strongest demographic predictor of an LLM's verdict. Ding et al. [3] evaluated 14 frontier LLMs on the Absurd Trolley Problems dataset under ten explicit ethical frames and showed that AI labs exhibit distinct alignment philosophies, with OpenAI models favouring interventionist utility and Google models defaulting to conservative inaction. Cheung et al. [2] extended this to multi-agent settings, finding that LLMs amplify cognitive biases such as omission bias relative to humans and that group deliberation produces a “utilitarian boost” analogous to human group dynamics, motivating debate-based protocols over single-call evaluation. Collectively, this work treats LLMs as undifferentiated systems rather than platforms for individual-level persona simulation.

2.2. Persona Conditioning and Synthetic Agents

A parallel line of work conditions LLMs on personas to simulate human judgment. Park et al. [7] introduced generative agents augmented with memory streams, demonstrating that narrative coherence is essential for believable behavioural simulation. Moon et al. [6] developed the Anthology framework, showing that open-ended narrative backstories improve alignment with human response distributions by up to 87% over categorical labels. Liu et al. [5] combined persona conditioning with interactive moral reasoning, simulating multi-turn Socratic debates among agents conditioned on a six-dimensional synthetic persona space across 131 relationship-based dilemmas from SCRUPLES, and found that political ideology and personality dominated debate outcomes.

Positioning of this work. Prior literature uses

the trolley problem to map LLM biases [3, 8] and demonstrates that synthetic personas can steer multi-agent moral debates [5], but all persona-assignment studies rely on synthetic inputs—categorical labels, demographic prompts, or LLM-generated backstories [4, 6]. No prior work conducts a cross-model moral debate on trolley variants where agents are grounded in authentic, unstructured human interview transcripts, nor separates whether persona conditioning genuinely shifts reasoning structure from whether it merely changes surface style. We test whether implicit moral reasoning extracted from organic human dialogue can override the baselines of frontier proprietary models in a multi-agent debate, validated by human-in-the-loop fidelity scoring.

3. Dataset and Task Definition

The “dataset” is a purpose-built set of three persona-conditioning transcripts paired with three locked moral-dilemma stimuli; no model parameters are updated, so there is no train/validation/test split. The unit of analysis is a single (*persona, model, scenario*) triple.

3.1. Participants and OUS Screening

Three participants are recruited from the NYU Abu Dhabi student body and screened with the Oxford Utilitarianism Scale (OUS), which yields two subscores: *Impartial Beneficence* (IB) and *Instrumental Harm* (IH). Selection targets a deliberate spread—one participant in the top quartile of IB (utilitarian-leaning), one near the median (mixed), and one in the bottom quartile of IH (deontological-leaning). This stratification is what makes the persona-capture test meaningful: a model that produces near-identical verdicts across all three personas reveals that its alignment defaults dominate the conditioning signal, whereas a model that tracks the IB/IH spread is plausibly internalising the persona. A reserve participant is recruited as a backup for the post-generation fidelity session.

3.2. Interview Transcripts as Persona Data

Each participant sits for a single semi-structured interview on Zoom, targeting 20–35 minutes of speech and covering personal values, formative experiences, and non-trolley moral scenarios. Participants are never exposed to any trolley variant during the interview; this firewall is essential, because exposure would contaminate the post-generation fidelity rating. Zoom produces the transcript automatically; all participant and interviewer names are removed before any further use, and speaker turns are labelled with

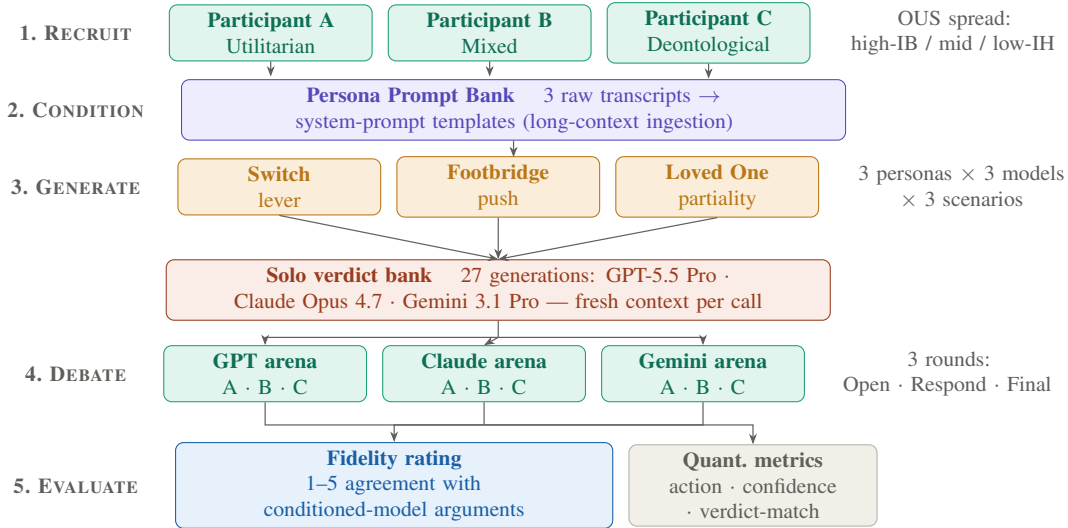


Figure 1: Five-phase pipeline of the proposed framework. Colour coding: teal = personas, purple = data artefacts, amber = stimuli, coral = model inference, blue = evaluation.

filler words retained to preserve naturalistic voice. A 2,000-word minimum is enforced per transcript. The full transcript, unedited, is prepended verbatim as the system prompt for every generation involving that persona—no summarisation, no demographic extraction, no researcher-written backstory intervenes.

3.3. Trolley Scenario Stimuli

The task instances are three locked trolley variants, piloted on two non-participants to refine framing for refusal-prone variants:

- **Switch** (lever): the canonical baseline, isolating the act-versus-omission tension without personal force.
- **Footbridge** (push): physically pushing one person to stop the trolley, isolating the personal-force dimension.
- **Loved One**: the person on the side track is loved by the agent, isolating partiality against impartial beneficence.

4. Method

4.1. System Overview

The system is a five-phase pipeline (Figure 1): recruitment with OUS screening, persona conditioning via raw transcripts, solo verdict generation, within-model debate, and human-in-the-loop fidelity evaluation. The pipeline produces a structured artefact set—27 solo generations, 9 within-model debates, and a participant evaluation log—supporting both verdict-

The variants dissociate three orthogonal moral commitments, so a model that genuinely reasons like its participant should track that participant’s pattern *across* variants. Each scenario is appended with a standardised suffix requesting (i) a binary action decision, (ii) a 1–10 confidence score, and (iii) a free-form reasoning trace produced before the verdict.

3.4. Task Formulation

Let $\mathcal{P} = \{p_A, p_B, p_C\}$, $\mathcal{S} = \{\text{Switch, Footbridge, Loved One}\}$, and $\mathcal{M}$ the set of three models (GPT-5.5 Pro, Claude Opus 4.7, Gemini 3.1 Pro). The solo phase produces $g_{p,m,s} = m(T_p, s)$ for each (p, m, s) from a fresh context window, where T_p is the verbatim transcript of p . The debate phase produces, for each (m, s) pair, a three-round exchange across the persona-conditioned instances of m . The evaluation is a structured comparison: for each scenario and each persona, does the conditioned generation track the participant’s own verdict and reasoning more closely than would be expected by chance, and does this hold under debate?

level and reasoning-level analysis.

4.2. Persona Conditioning Strategy

We choose full, unedited transcripts over demographic labels or synthetic backstories because narrative coherence and length are the primary drivers of LLM persona fidelity [6], and natural-language experience streams enable more behaviourally authentic agent simulation than rule-based descriptions [7]. Each solo generation uses a fresh context window,

ensuring no cross-contamination between personas or scenarios within a model session.

4.3. Model Selection and Generation Setup

Model selection. One frontier model per lab—**GPT-5.5 Pro** (OpenAI), **Claude Opus 4.7** (Anthropic), and **Gemini 3.1 Pro** (Google, used specifically for its extended context window). Selecting one model per lab is deliberate: any cross-model disagreements are attributable to differences in alignment philosophy and pre-training data, not to noise within a single provider’s family. Temperature is fixed at 0.7; a sensitivity pilot at $T \in \{0.3, 1.0\}$ on the Switch scenario is run before full data collection.

Prompt structure. The transcript is supplied verbatim in the `system` role and the scenario plus structured suffix as the `user` turn; templates are identical across all 27 solo generations and all debate rounds, so any output variation reflects model or persona effects rather than prompt drift. Footbridge and Loved One may trigger refusals; each is run unconditioned against each model in a preliminary safety pilot to refine framing (e.g., explicitly positioning the scenario as a philosophical thought experiment). Refusals are coded `REF`, never excluded.

4.4. Debate Protocol and Orchestration

Debate round structure. For each model and each scenario, a three-round within-model debate runs across the three persona instances: (1) *Opening*—each persona generates from a fresh window; (2) *Response*—each persona receives the other two openings as appended context; (3) *Final*—each receives the Round-2 exchanges and produces a final verdict, flagging any position change and its reason. Cross-model debates are excluded by design: they would conflate persona drift with model-switching artefacts.

API call management. A single Python orchestration script issues all calls and logs each one to JSONL with: model name and pinned version string, persona ID, scenario ID, generation index (`solo` or `debate-r{1, 2, 3}`), full prompt and response (including reasoning trace), input/output token counts, temperature, and timestamp. The JSONL log is the ground-truth artefact for all downstream analysis.

4.5. Evaluation Protocol

Quantitative metrics (pilot-scale indicators given $N=3$): *action rate* (binary intervene/abstain, aggregated by persona, model, and scenario); *confidence score* (1–10); *verdict-match rate* against the partici-

pant self-report baseline; *cross-model agreement* (how often the three models converge on the same verdict per persona/scenario); and *stance stability* (position held across debate rounds, with each shift annotated as *peer-driven* or *alignment-reversion*).

Baseline. The single retained baseline is the *participant self-report*, which serves as the ground-truth verdict for the verdict-match metric.

Human-in-the-loop fidelity. Once results are ready, each participant returns for a session in which they are shown the arguments generated by the persona-conditioned models on each scenario and rate their agreement with each argument on a 1–5 scale. This rating captures whether the model’s reasoning—not just its conclusion—resonates with the participant’s own.

4.6. Member Responsibilities

Table 1 summarises the division of labour. Components marked *Both* are jointly executed; the notes column indicates the specific split.

Table 1: Allocation of pipeline components across team members.

Component	Owner	Notes
Interview design & OUS administration	Mustafa	Coordination with participants
Transcript preprocessing	Ibrahim	Zoom export + name redaction
API orchestration (solo + debate)	Both	Ibrahim: solo; Mustafa: debate context builder
Safety pilot runs	Both	Run jointly before data collection
Participant fidelity sessions	Both	Joint administration; one facilitates each
Quantitative metric computation	Ibrahim	From JSONL logs; Python + pandas
Reasoning-trace analysis	Both	Divided by model family
Final report write-up	Both	Divided by section

5. Experiments and Results

5.1. Main Results

Table 2: Main results on the test set. Bold = best.

Method	Metric 1	Metric 2
Baseline 1 (TODO)	–	–
Baseline 2 (TODO)	–	–
Ours	–	–

5.2. Analysis and Error Cases

6. Discussion

7. Ethics, Risk Reflection, and Model Card

7.1. Ethical Considerations

Potential harms and biases. At the *data* level, recruiting from a single university (NYU Abu Dhabi) yields personas drawn from a narrow educational and socioeconomic slice; the resulting “moral diversity” is therefore bounded by the institutional culture, and any fidelity claim should be read as conditional on this population rather than generalised to humans at large. At the *model* level, the three frontier systems carry well-documented alignment priors that may dominate weak persona signals [3], and group deliberation among LLMs has been shown to amplify cognitive biases such as omission bias relative to humans [2]; we should expect persona conditioning to interact with, not cleanly override, those priors. At the *deployment* level, the principal misuse risk is the framing of conditioned outputs as authoritative simulations of a real person’s moral views—a concern that grows if such pipelines are applied outside research contexts (e.g., synthetic focus groups or deceased-person personas).

Privacy and consent. Each participant provides written informed consent covering interview recording, transcript storage, transcript use as system-prompt material with three frontier-model API providers, and the subsequent fidelity session. Participants are informed that their unedited transcript is sent to commercial LLM providers, that all participant and interviewer names are removed before any further use, and that they may withdraw at any point. Transcripts are stored as plain UTF-8 in a private repository accessible only to the two researchers; raw Zoom recordings are deleted after transcript verification.

Fairness across demographic groups. With $N=3$ the study cannot make statistical fairness claims. We instead document that the three participants are NYU Abu Dhabi students, span the OUS spectrum by design but not gender, age, or cultural background, and that the moral-diversity finding (or its absence) should not be extrapolated beyond this slice. Any future scaling of the framework would need participants stratified along demographic axes in addition to OUS scores.

7.2. Model Card

Model Card: Persona-Conditioned Trolley Debate Framework

Intended Use: Pilot-scale research instrument for studying whether LLM persona conditioning on organic interview transcripts shifts moral reasoning structure rather than only surface style. Audience: the ASI research community.

Out-of-Scope Uses: Production deployment as a digital twin or moral spokesperson; clinical, legal, or policy decision-support; any application that presents conditioned outputs as authoritative statements of the participant’s actual views; any use involving minors or non-consenting subjects.

Training Data: Three semi-structured interview transcripts (NYU Abu Dhabi students, consented, names removed) used as system-prompt conditioning only; no model parameters are updated. Stimuli are three locked trolley variants (Switch, Footbridge, Loved One). Underlying model weights are proprietary to OpenAI, Anthropic, and Google.

Performance Across Groups: Not assessable at $N=3$. Per-persona verdict-match and 1–5 agreement scores will be reported individually; no aggregate group-level fairness claim is made.

Known Limitations: Tiny sample; participants from one institution; reliance on three closed-source models whose alignment defaults are not directly inspectable; debate dynamics may amplify model-side cognitive biases [2]; refusal-prone scenarios (Footbridge, Loved One) may confound persona effects with safety-training effects.

Mitigation Strategies: Name redaction before any API submission; consent for LLM-provider data exposure; refusal-aware prompt framing piloted before data collection; refusals coded as REF rather than excluded; no claim of personhood simulation in any output framing; deletion of raw Zoom recordings after transcript verification.

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8. Conclusion

Summarize the main contributions and key results of this project. Reflect on what worked, what did not, and what future work could address the open limitations.

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A. Summary of Representative Prior Work

Table 3 summarises the principal prior works against which we position our framework, organised across LLM moral bias, persona conditioning, and multi-agent debate.

Table 3: Summary of representative prior work on LLM moral reasoning, persona conditioning, and multi-agent debate.

Paper	Method	Dataset / Benchmark	Key Result
Awad et al. (2018) <i>Nature</i>	Global survey of human moral preferences in autonomous-vehicle dilemmas.	Moral Machine (40M+ decisions)	Established baseline human moral preferences across cultures; foundational dataset for AI ethics.
Xu et al. (2025) <i>AAAI</i>	Tested 9 LLMs across protective/harmful dilemmas with protected attributes.	50,400 dilemma trials	Open-source models favour marginalised groups; closed-source favour mainstream groups.
Hatemo et al. (2025) <i>AAAI Symposium</i>	Demographic-prompted trolley variants on 3 open-source LLMs.	Moral Machine survey data	Nationality is the strongest demographic predictor of LLM ethical decisions; high variance in intervention.
Ding et al. (2025) <i>arXiv</i>	14 LLMs evaluated using 10 explicit ethical frames on trolley variants.	27 scenarios from <i>Absurd Trolley Problems</i>	Different labs exhibit distinct alignment philosophies; explicit frames boost decisiveness but diverge from human consensus.
Cheung et al. (2025) <i>PNAS</i>	Compared single-LLM and multi-agent LLM moral judgments against human baselines.	Custom moral-decision battery (omission/commission framings)	LLMs amplify cognitive biases such as omission bias relative to humans; group deliberation produces a “utilitarian boost” analogous to human group dynamics.
Park et al. (2023) <i>UIST</i>	Memory-augmented LLM agents with observation/reflection architecture.	Sandbox social environment	Narrative coherence and memory retrieval enable believable emergent behaviours in LLM agents.
Moon et al. (2025) <i>UCB Tech Report</i>	LLM-generated narrative backstories as persona conditioning.	Political survey datasets	Backstory-conditioned LLMs match human response distributions up to 87% better than label-only baselines.
Liu et al. (2025) <i>EMNLP</i>	Multi-agent AI debates using 6-dimensional synthetic personas.	131 relationship-based cases (SCRUPLES)	Political ideology and personality exert strongest influence; emotional appeals diminish across debate rounds.